SIMATS ENGINEERING

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Shaik Rehan
192525110RC

Class Work(28.07.26)

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QUESTIONS

Q1
Keyword Arguments
5 marks

Q2
Keyword Arguments
5 marks

Q3
Variable-Length Arguments (*args)
5 marks

Q4
Variable-Length Arguments (*args)
5 marks

Q5
Variable-Length Arguments (*args)
5 marks

Q6
Variable-Length Arguments

Q1 **Keyword Arguments** 5 marks

Write a function employee(empid, name, salary) and call the function by changing the order of the arguments using keywords.

Your Code

```
def employee(empid, name, salary):
    print("Employee ID:", empid)
    print("Name:", name)
    print("Salary:", salary)

employee(salary=50000, name="Rehan", empid=101)
```

Input

stdin input

Output

```
Employee ID: 101
Name: Rehan
Salary: 50000
```

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- **Q5** Variable-Length Arguments (*args) 5 marks
- **Q6** Variable-Length Arguments

Q2 Keyword Arguments

5 marks

Write a function book(title, author, price) and display the book details using keyword arguments.

Your Code

```python
def book(title, author, price):
    print("Book Details")
    print("Title :", title)
    print("Author:", author)
    print("Price :", price)
title = input("Enter title: ")
author = input("Enter author: ")
price = float(input("Enter price: "))
book(title=title, author=author, price=price)
```

Input

```
the alchemist
paulo coelho
499
```

Output

```
Enter title: Enter author: Enter
price: Book Details
Title : the alchemist
Author: paulo coelho
Price : 499.0
```

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Variable-Length Arguments

Q3 **Variable-Length Arguments (*args)** 5 marks

Write a function that accepts any number of integers and returns their sum.

Your Code

```python
def add_numbers(*args):
    return sum(args)
print(add_numbers(1, 2, 3, 4))
```

Input

```
1 2 3 4
```

Output

```
10
```

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Q6
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Q4 **Variable-Length Arguments (*args)** 5 marks

Write a function that accepts any number of marks and returns the average.

Your Code

```
def average_marks(*marks):
    if len(marks) == 0:
        return 0
    return sum(marks) / len(marks)

marks = list(map(int, input().split()))
print(average_marks(*marks))
```

Input

```
3 5 7 9
```

Output

```
6.0
```

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Variable-Length Arguments (*args)
5 marks

Q6
Variable-Length Arguments

Q5 **Variable-Length Arguments (*args)** 5 marks

Write a function that accepts any number of strings and prints each string on a new line.

Your Code

```
def print_strings(*strings):
    for s in strings:
        print(s)

n = int(input())
strings = []
for _ in range(n):
    strings.append(input().strip())

print_strings(*strings)
```

Input

```
3
1
2
3
```

Output

```
1
2
3
```

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5 marks

Q6
Variable-Length Arguments

Q6 **Variable-Length Arguments (*args)** 5 marks

Write a function that accepts any number of numbers and returns the largest value.

Your Code

```
def largest(*args):
    return max(args)

nums = list(map(int, input().split()))
print(largest(*nums))
```

Input

```
7 2 9 4 5 6
```

Output

```
9
```

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5 marks

Q6
Variable-Length Arguments

Q7 **Recursive Functions** 5 marks

Write a recursive function to generate the Fibonacci series up to n terms.

Your Code

```python
def fib(n):
    if n == 0:
        return 0
    if n == 1:
        return 1
    return fib(n - 1) + fib(n - 2)
def fibonacci_series(n):
    series = []
    for i in range(n):
        series.append(fib(i))
    return series
n = int(input("Enter n: "))
print(fibonacci_series(n))
```

Input

```
7
```

Output

```
Enter n: [0, 1, 1, 2, 3, 5, 8]
```

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5 marks

Q6
Variable-Length Arguments

Q8 **Recursive Functions**

5 marks

Write a recursive function to find the sum of digits of a number.

Your Code

```
def sum_of_digits(n):
    n = abs(n)
    if n < 10:
        return n
    return (n % 10) + sum_of_digits(n // 10)
num = int(input())
print(sum_of_digits(num))
```

Input

```
1001
```

Output

```
2
```

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5 marks

Q6
Variable-Length Arguments

Q9 **Recursive Functions**

5 marks

Write a recursive function to reverse a string.

Your Code

```
def reverse_string(s):
    if len(s) <= 1:
        return s
    return reverse_string(s[1:]) + s[0]
text = "hello"
print(reverse_string(text))
```

Input

```
Hello
```

Output

```
olleh
```

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Q5
Variable-Length Arguments (*args)
5 marks

Q6

Q10 **Recursive Functions** 5 marks

Write a recursive function to calculate x? (power of a number).

Your Code

```
def power(x, n):
    if n == 0:
        return 1
    if n < 0:
        return 1 / power(x, -n)
    return x * power(x, n - 1)

# Example usage:
x = float(input("Enter x: "))
n = int(input("Enter n: "))
print(power(x, n))
```

Input

```
2
-2
```

Output

```
Enter x: Enter n: 0.25
```

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